Guidelines for the Description of Soybean Germplasm Resources

1 Scope

This specification establishes the descriptors for soybean germplasm resources and their grading criteria.

These guidelines apply to the collection, organization, and preservation of soybean germplasm resources; the development of data standards and data quality control specifications; and the establishment of database and information-sharing network systems.

2 Normative References

The provisions in the following documents shall form part of this specification by reference therein. For dated reference documents, all subsequent corrigenda (excluding errata) or revised editions shall not apply to this specification. However, parties to agreements concluded in accordance with this specification are encouraged to consider whether the latest editions of these documents may be used. For undated reference documents, the latest edition shall apply to this specification.

ISO 3166 Codes for the Representation of Names of Countries

GB/T 2659: Codes for Place Names Worldwide and by Region

GB/T 2260: Administrative Division Codes of the People's Republic of China

GB/T 12404 Code for Unit Affiliation

GB/T 1.1−2002 Guidelines for Standardization Work − Part 1: Structure and Drafting Rules for Standards

GB/T 3543.6-1995 Code for Inspection of Crop Seeds

GB 4404.2-1996 Grain seeds − Legumes

3 Terms and Definitions

3.1 Soybean

Leguminosae, Papilionoideae, and Glycine. Within the genus Glycine, there are two subgenera: Glycine and Soja. The Glycine subgenus comprises 16 species (Glycine *albicans; G.arenaria;G.argyrea;G.canescens;G.clandestina;G.curvata;G.cyrtoloba; G.falcata;G.hirticaulis;G.lactovrirens;G.latifolia;G.latrobeana;G. microphylla; G. pindanica; G. tabacina; G. tomentella).* The genus G. soja comprises two species: the annual wild soybean (G. soja Sieb. et Zucc.) and the cultivated soybean (G. *max* (L.) Merr.). The seeds of cultivated soybean contain approximately 40% protein and 20% fat; it serves not only as an important food crop and a primary oilseed crop, but also as livestock feed and a raw material for food and light industry, thereby occupying a significant position in the national economy.

3.2 Soybean Germplasm Resources

Local soybean varieties, selected and bred varieties, lines, imported foreign materials, wild soybean resources, and genetic materials, among others.

3.3 Basic Information

Basic descriptive information on soybean germplasm resources, including the national unified identification number, germplasm name, scientific name, place of origin, germplasm type, and other details.

3.4 Morphological Characteristics and Biological Properties

Phenological stages, botanical morphology, yield traits, and other characteristic traits of soybean germplasm resources.

3.5 Quality Characteristic

The crude protein content, crude fat content, amino acid content, fatty acid content, isoflavone content, and other parameters of soybean germplasm resources.

3.6 Stress Resistance

The adaptability or resistance of soybean germplasm resources to various abiotic stresses, including cold tolerance, drought tolerance, and salt tolerance.

3.7 Resistance to Pests and Diseases

The adaptability or resistance of soybean germplasm resources to various biological stresses, including soybean mosaic virus disease, soybean downy mildew, and soybean cyst nematode disease.

4 Essential Information

4.1 Nationally Unified Numbering

The sole identifier for germplasm: the unified domestic numbering for cultivated soybean germplasm resources consists of "ZDD" followed by a 5-digit serial number; the unified numbering for foreign-introduced cultivated soybean germplasm consists of "WDD" followed by a 5-digit serial number; the unified numbering for wild soybeans consists of "ZYD" followed by a 5-digit serial number. The unified numbering for food soybeans preserved by the Institute of Vegetable and Flower Research of the Chinese Academy of Agricultural Sciences consists of "V07F" followed by a 4-digit serial number.

4.2 Germplasm Bank Identification Number

The numbering system for soybean germplasm in the national long-term crop germplasm resource repository. The numbering for cultivated soybean long-term repositories consists of "I2A" followed by a 5-digit sequence number; for wild soybean long-term repositories, it consists of "I2B" followed by a 5-digit sequence number; and for the edible soybean long-term repository preserved by the Institute of Vegetable and Flower Research of the Chinese Academy of Agricultural Sciences, it consists of "*II*7F" followed by a 4-digit sequence number.

4.3 Introduction Number

The designation assigned to soybean germplasm when it is introduced from abroad.

4. 4 Collection Number

The identification number assigned to soybean germplasm during field collection

4.5 Germplasm Designation

The Chinese name for soybean germplasm.

4.6 Foreign Name of Germplasm

The foreign-language name of the introduced germplasm or the Chinese Pinyin name of the domestic germplasm.

4. 7 Subject Names

Leguminosae.

4. 8 Genus Name

The soybean genus (Glycine*)*.

4.9 Scientific Name

The scientific name for soybean is Glycine max (L.) Merrill; the annual wild soybean is G. soja Sieb. et Zucc.

4.10 Country of Origin

The name of the country of origin, the name of the region, or the name of the international organization of the soybean germplasm.

4.11 Country of Origin

Name of the province of origin for domestic soybean germplasm; name of the national first-level administrative region of origin for germplasm introduced from abroad.

4.12 Country of Origin

The names of the original counties, townships, and villages of domestic soybean germplasm.

4.13 Height

Altitude of the original habitat of the soybean germplasm. Unit: m.

4. 14 Longitude

The longitude of the origin of soybean germplasm is expressed in degrees (°) and minutes (′). The format is DDDFF, where DDD represents degrees and FF represents minutes. East longitude is positive, while west longitude is negative.

4.15 Latitude

The latitude of the original origin of soybean germplasm is expressed in degrees (°) and minutes (′), formatted as DDFF, where DD represents degrees and FF represents minutes. North latitude is a positive value, while south latitude is a negative value.

4. 16 Place of Origin

For soybean germplasm introduced from abroad, provide the name of the country, region, or international organization of origin; for domestic germplasm, provide the name of the province or county of origin.

4.17 Preserving Entity

The name of the original institution where the soybean germplasm was preserved prior to its submission to the National Long-term Crop Germplasm Resource Bank.

4.18 Storage Unit Number

The germplasm accession number assigned by the original germplasm preservation institution.

4.19 Genealogy

The genetic relationships among soybean breeding varieties (lines).

4.20 Breeding Unit

Name of the institution or individual responsible for the breeding and development of soybean varieties (lines).

4.21 Age at Maturation

Year of breeding and development of the soybean variety (line).

4.22 Breeding Method

Methods for breeding soybean varieties (lines).

4.23 Germplasm Types

Soybean germplasm types are divided into six categories.

1 Annual wild species

2 Perennial wild species

3 local variety

4 Selected breeding varieties

5 Product line

6 genetic stocks

7 other

4.24 Utilization Type

The utilization types of soybean germplasm are divided into three categories.

1 For granules

2 For cooking

3 For feed

4.25 Sowing Type

Soybean germplasm sowing periods are divided into four categories.

1 spring

2 Xia

3 Autumn

4	winter

4.26 Ecological Zone

Based on the light and temperature conditions of the origin of soybean germplasm, soybean germplasm ecological zones are classified into three categories.

1	Northern region
2	Huang-Huai-Hai Region
3	south

4.27 Reproductive Period Group

Based on the response of Chinese soybean germplasm resources to light and temperature, soybean germplasm resources are divided into 12 growth stage groups.

1	**000**
2	**00**
3	0
4	I
5	II
6	III
7	IV
8	V
9	VI
10	VI
11	VIII
12	IX

4.28 picture

Image filename for soybean germplasm. Image format:.jpg.

4.29 Observation Site

Name of the observation site for the morphological characteristics and biological traits of soybean germplasm.

5 Morphological Characteristics and Biological Properties

5.1 Design and Colour

The color of the petals on the day of flowering.

1	white
2	purple

5.2 Length of the Inflorescence

The length from the inflorescence attachment point to the apex during the initial pod-setting stage.

1 short

2 China

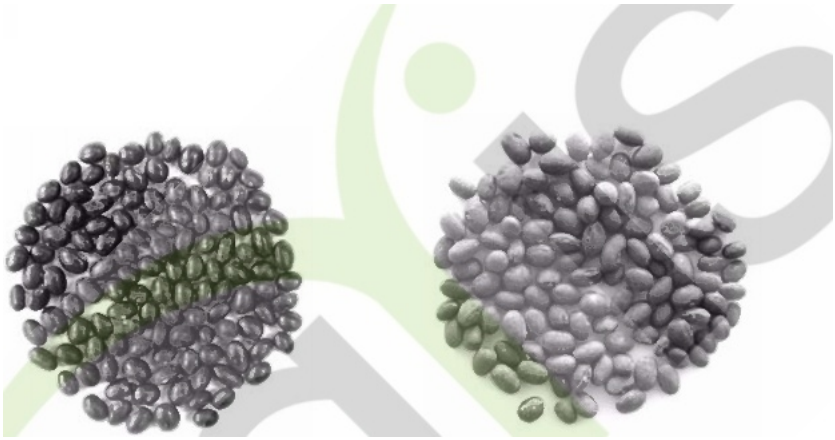
3 long

5.3 Clay mask

Presence or absence of seed mucus in normally matured seeds (see Figure 1).

0 not have

1 have



0

Figure 1: Seed Mud Film

5.4 Granule Color

The color of the seed coat of normally mature seeds.

1 Huang

2 Green

3 black

4 brown

5 two-tone

5.5 Wild Soybean Grain Color

The color of the seed coat in normally mature wild soybean seeds.

1 Huang

2 Green

3 black

4 brown

5 two-tone

5.6 Seed Coat Luster

The luster of the seed coat of mature seeds.

0 not have

1 minute

2 Strong

5.7 Morphology

The shape of mature seeds.

1 circle

2 Flat and round

3 oval

4 Flat oval

5 Long oval

6 Reniform

5.8 Grain size

Based on the weight of 100 mature seeds, seed size is classified into five grades.

1 Extremely small

2 small

3 China

4 big

5 especially big

5.9. Epidermal Fissures

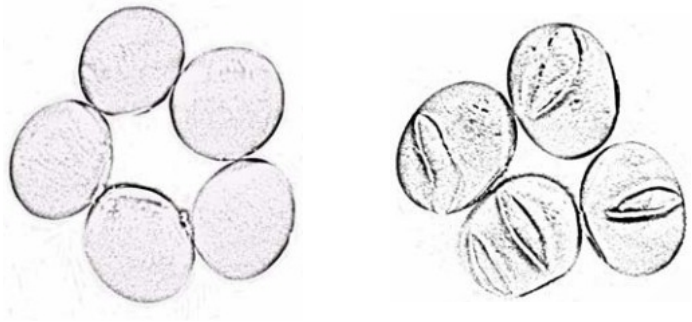
Natural cracking degree of the epidermis of mature seeds (see Figure 2).

1 No cracking

3 light

5 China

7 Prone to cracking



1 The testa does not split 2. Pericarpic splitting

Figure 2: Seed cracks

5.10 Cotyledon Color

Color of the cotyledons in mature seeds.

1 Huang

2 Green

5.11 Navel Color

Color of the hilum of mature seeds.

1 Huang

2 Light brown

3 brown

4 Deep brown

5 Lan

6 Light black

7 black

5.12 Fur color

The color of the trichomes on the upper part of the stem and the pod skin when mature.

1 Gray

2 Brown

5.13 Fuzz Density

The density of the pubescence in the upper-middle part of the stem when mature.

0 not have

1 rare

2 China

3 secret

5.14 Height of the Pile

The growth status of trichomes on mature leaves during the peak flowering stage.

1 up-right

2 bias

3 Close to

5.15 Pod color

The color of the pod when mature.

1 Grayish-brown

2 Chloasma

3 brown

4 Deep brown

5 black

5.16 Petiole Length

Based on the length of the petiole of mature leaves, petiole length is divided into two categories (see Figure 3).

1 short

2 long

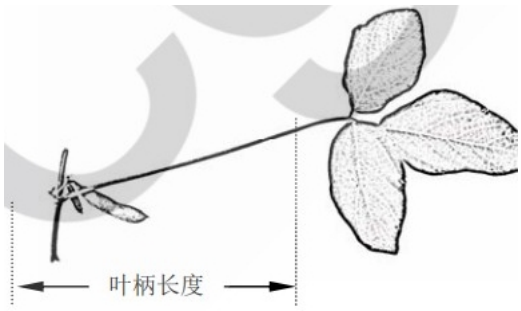


Figure 3: Petiole Length

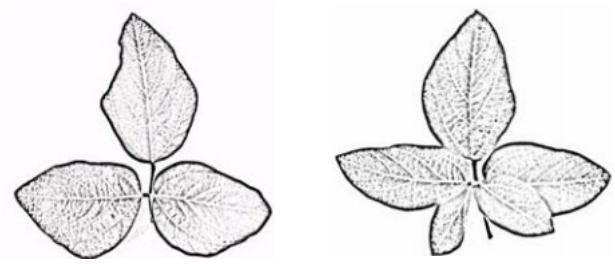
5.17 Number of Leaflets

During the peak flowering stage, the number of mature compound leaflets in the middle and upper parts of the plant (see Figure 4).

1 3

2 4–6

3 No fewer than 7



1 2

Figure 4: Number of Leaflets

5.18 Leaf

The shape of the terminal leaflet at the apex of the fully developed trifoliate leaves in the upper-middle part of the plant during the peak flowering stage (see Figure 5).

1 Lanceolate

2 Oval

4 oval

4 circle

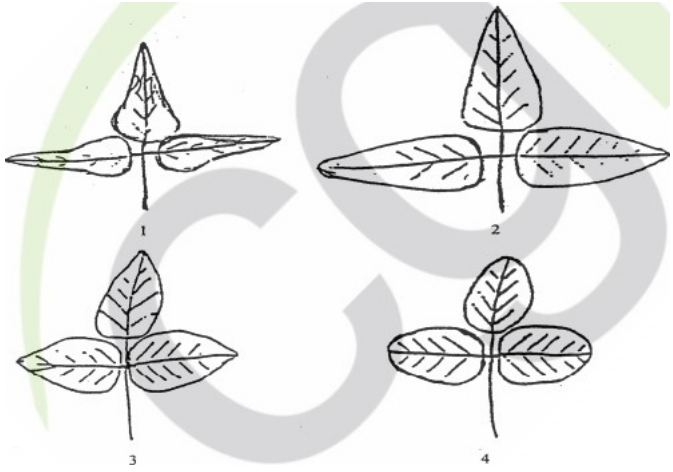


Figure 5: Leaflet Shape

5. 19 Leaf Color

During the peak flowering stage, the color of the leaves in the middle and upper parts of the plant.

1 acid green

2 Green

3 dark green

5.20 Leaf size

Based on the area of the apical leaflet of the fully developed trifoliate leaves in the upper-middle part of the plant during the peak flowering period, leaflet size was classified into three categories.

1 small

2 China

3 big

5.21 Deciduousness

The degree of leaf shedding when the plant reaches maturity and is harvested.

1 Do not fail to

2 Half-set

3 leave out

5.22 Growth habits

The growth morphology of the main stem during the middle and late stages of growth and development.

1 up-right

3 semi-erect

5 semi-hydrarpodous

7 rampancy

5.23 Podding habit

The pod formation status of the plant at maturity (see Figure 6).

3 infinite

5 Yahou Limited

7 limited



3 5 7

Figure 6: Podding Habit

5.24-Unit Height

Height from the cotyledon node to the plant growth point when mature. Units: cm (see Figure 7).

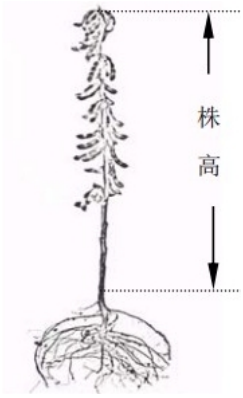


Figure 7: Plant Height

5.25 Effective Branch Count

At maturity, the number of first-order branches on the plant that bear two or more stem nodes and one or more mature pods; expressed in units (see

Figure 8).

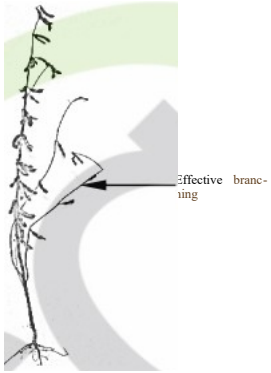


Figure 8: Number of Effective Branches

5.26 Strain Type

The degree of convergence between the mature lateral branches and the main stem (see Figure 9).

3 convergence

5 Half-open

7 open a business

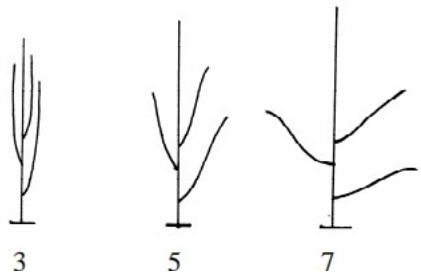


Figure 9: Plant Habit

5.27 Basal Stem Color

The color of the hypocotyl when the first trifoliate leaf of the plant unfolds.

1 Green

2 purple

5.28 Main Stem of Soybean

During the peak flowering period of wild soybeans, the prominence of the main stem relative to the branches is more pronounced.

1 clear

2 Not displayed

5.29 Number of Nodes on the Main Stem

The number of nodes from the cotyledon node to the tip of the main stem when the plant has reached maturity. The unit is the number of nodes (see Figure 10).

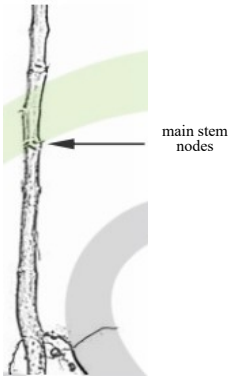


Figure 10: Main stem nodes

5.30 Stem Diameter

The diameter of the fifth internode of the main stem when the plant has reached maturity. Units: mm (Figure 11).



Figure 11: Stem Diameter

5.31 Stem Shape

The shape of the main stem when the plant has matured (see Figure 12).

1 Normal stem

2 flat stem

3 Curved stem



1 2 3

Figure 12: Main Stem Shape

5.32 Straw Stiffness

The toughness of the stem during the later stages of plant growth and development.

1 weak

2 China

3 Strong

5.33 Lodging

The degree of plant tilt during the later stages of plant growth and development.

1 Does not topple

3 Lightly pour

5 Mid-inversion

7 Heavy inversion

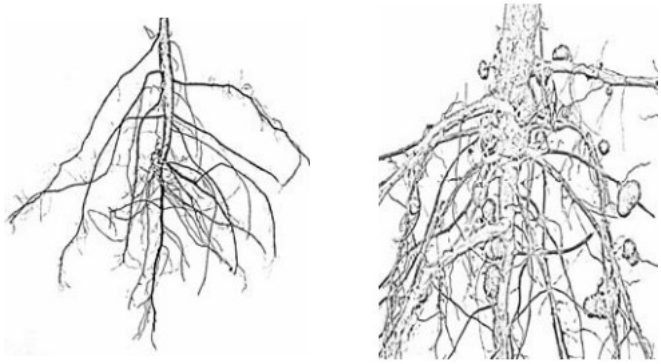
9 Severe inversion

5.34 Rhizoma

Presence or absence of root nodules at the initial flowering stage (see Figure 13).

0 not have

1 have



0 1

Figure 13: Rhizome

5.35 Number of Pods per Plant

The actual number of pods produced per plant at maturity. The unit is number of pods.

5.36500g Number of Pods

When harvesting fresh soybean pods for use in cooking, the number of fresh pods contained in a 500 g pod weight. The unit is number of pods.

5.37 Pod Size

When maturity is reached, the pod size is classified into three categories based on the length of the pod.

1 small

2 China

3 big

5.38 Capsule length

Vegetable: length of fresh soybean pods at harvest. Unit: cm (see Figure 1). 4 ）。

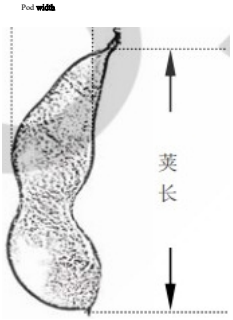


Figure 14: Pod Length and Pod Width

5.39 Pod Width

The width at the widest point of the fresh soybean pods when they are harvested. The unit is cm (see Figure 14).

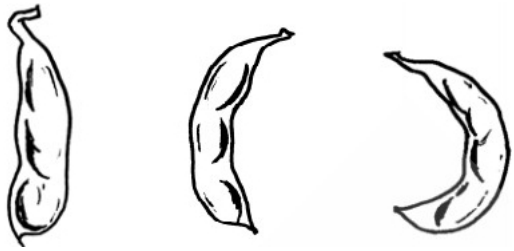
5. 40 Pod-Shaped

The shape of the pods in the middle to upper part of the main stem from the peak growth stage through maturity (see Figure 15).

1 Rectangular shape

2 Sickle-shaped

3 segment of a circle



1 2 3

Figure 15: Shape of the Legume

5.41 Pod Height

At maturity, the height from the cotyledon node to the lowest pod, to the site where the flowering inflorescence or first effective branch is borne. Unit: cm.

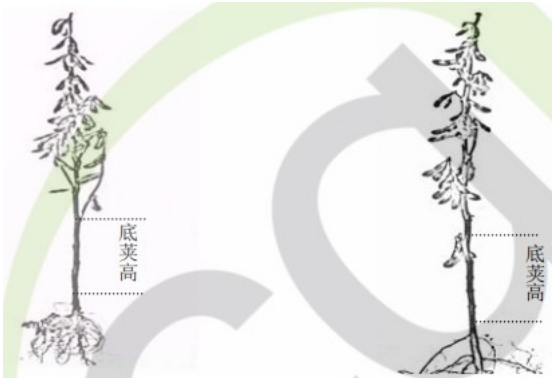


Figure 16: Pod Height

5.42 Fissured Pod

The proportion and grade of natural pod splitting upon maturity.

3 No cracking

5 Minor crack

7 China

9 Prone to cracking

5.43 Number of grains per plant

At maturity, the actual number of grains per plant, excluding empty grains and including immature grains, diseased grains, and insect-infested grains; expressed in grains.

5.44 Seeds per Pod

Average number of grains per plant at maturity. Unit: grains.

5.45 Single-Grain Weight

The actual weight of seeds harvested per plant at maturity. Unit: g.

5.46 Fresh Hundred-Grain Weight

The weight of 100 intact grains when fresh soybean pods are harvested. Unit: g.

5. 47 Hundred-Grain Weight

Weight of 100 intact grains after maturity. Unit: g.

5.48 Yield per Unit Area

Net weight of harvested seeds per unit area at maturity, expressed in kg/ha².

5.49 Pregnancy Month

The number of months experienced during the growth period of soybean germplasm.

5.50 Number of Growing Days

For spring soybeans, the number of days from the day after emergence to the day of maturity; for summer and autumn soybeans, the number of days from the day after sowing to the day of maturity. The unit is d.

5.51 Sowing Time

For the identification of morphological characteristics and biological traits of soybean germplasm resources, the seed sowing date shall be expressed in the format "YYYYMMDD," using "year month day."

5.52 Seedling Emergence Period

The date on which the number of seedlings emerging from the cotyledons in the residential community reaches 50% shall be expressed as "year month day" in the format "YYYYMMDD".

5.53 Flowering Period

The date on which the flowering rate in the residential community reaches 50% shall be expressed as "year month day" in the format "YYYYMMDD".

5.54 Podding Stage

The date on which 50% of the plants in the community develop pods exceeding 2 cm in length shall be indicated as "year month day," in the format "YYYYMMDD."

5.55 Granular Stage

For plots where 50% of the plant pods are flattened and the kernels are relatively prominently protruding, indicate the date in the format "YYYYMMDD".

5.56 Maturation Period

The date by which 95% of the pods on the entire plant have reached their mature color and produce a audible sound when shaken is recorded as "year month day" in the format "YYYYMMDD."

6 Quality Characteristics

6. 1 Crude protein content

Total protein content in the kernels, expressed as%.

6.2 Lysine Content

Lysine content in the kernels, expressed as%.

6.3 Tryptophan Content

Protein content in the grains, expressed as%.

6.4 Phenylalanine Content

Content of zylan phenylalanine, expressed as%

6.5 Methionine Content

Methionine content in the kernels, expressed as%.

6.6 Threonine Content

Threonine content in the kernels, expressed as%.

6.7 Isoleucine Content

Content of isoleucine in the kernels, expressed as%.

6.8 Leucine Content

Lecithin content in the kernels, expressed as%.

6.9 Valine Content

Content of lysine in the grains, expressed as%.

6.10 Cystine Content

Cystine content in the grains, expressed as%.

6.1111S/ 7S

The ratio of the 11S globulin content to the 7S globulin content in the grains.

6.12 Allergic Protein 28 KDa

Grain 28K: absence of allergen protein subunits.

0 not have

1 have

6.13 Allergen Protein 30 KDa

Grain 30K: Deficiency of allergen protein subunits.

0 not have

1 have

6.14 Kunitz **Tryptase Inhibitors** Seed; Deficiency

of Kunitz-Type Tryptase Inhibitors.

0 not have

1 have

6.15 Crude Fat Content

Fat content of the kernels, expressed as a percentage (%).

6.16 Stearic Acid Content

Content of saturated fatty acids in the kernels (18:0), expressed as%

6.17 Palmitic Acid Content

Content of saturated fatty acids in the kernels (16:0), expressed as%

6.18 Oleic Acid Content

The content of oleic acid (18:1) in the kernels, expressed as a percentage.

6.19 Linoleic Acid Content

The content of linoleic acid (18:2) in the kernels, expressed as%.

6.20 Linolenic Acid Content

The content of linolenic acid (18:3), expressed as%.

6.21 Fat Oxygenase

Deficiency of seed fat oxygenase Lox1, Lox2, and Lox3.

1 Lox1 deficiency

2 Lox2 deficiency

3 Lox3 deficiency

4 Lox1,2 deficiency

5 Lox1,3 deficiency

6 Lox2,3 deficiency

7 Deficiency in Lox1,2,3

8 No shortage of Lox

6.22 Sugar content

Sugars content in the grains, expressed as%.

6.23 Isoflavone Content

Content of seed isoflavones. The unit is 10^2 mg/g.

7 Stress Resistance

7.1 Salt Tolerance During the Budding Stage

The ability of soybean seedlings to tolerate salt stress during the germination stage.

1 be able to bear or endure
2 More durable
3 Zhongnai
4 Relatively sensitive
5 sensitive

7.2 Salt tolerance during the seedling stage

The ability of soybean seedlings to tolerate salt stress (primarily sodium chloride) during the seedling stage.

0 High durability
1 be able to bear or endure
2 More durable
3 Zhongnai
4 Relatively sensitive
5 sensitive

7. Salt tolerance during the 3-flower pod stage

The ability of soybean flowers and pods to tolerate salt stress (primarily sodium chloride).

1 be able to bear or endure
2 More durable
3 Zhongnai
4 Relatively sensitive
5 sensitive

7.4 Drought tolerance during the bud stage

The ability of soybeans to withstand drought stress during the germination stage.

1 be able to bear or endure
2 More durable
3 Zhongnai

4 Relatively sensitive

5 sensitive

7.5 Drought Tolerance of Mature Plants

The ability of soybeans to withstand drought stress under natural drought conditions.

1 be able to bear or endure

2 More durable

3 Zhongnai

4 Relatively sensitive

5 sensitive

7. Cold tolerance during the 6-bud stage

The ability of soybeans to withstand cold stress during the germination stage.

1 be able to bear or endure

2 More durable

3 Relatively sensitive

4 sensitive

7.7 Acid resistance and aluminum resistance

The ability of soybean plants to tolerate aluminum ion toxicity in acidic soils.

1 be able to bear or endure

2 sensitive

7.8 Acid Rain Resistance

The tolerance of soybean plants to acid rain stress.

1 be able to bear or endure

2 More durable

3 Zhongnai

4 Relatively sensitive

5 sensitive

8 Disease and Pest Resistance

8.1 Resistance to Soybean Rust

The resistance of soybean plants to Phakopsora pachyrhizi Sydow, the rust fungus of the bean tuber layer.

1 highly resistance(HR)

3 resist(R)

5 moderate resistance(MR)

7 feel(S)

9 High Sensitivity (HS)

8. 2 Resistance to soybean gray mold

The resistance of soybean plants to the gray spot fungus (Cercospora sojina Hara).

1 highly resistance(HR)

3 Anti-(R)

5 moderate resistance(MR)

7 feel(S)

9 High Sensitivity (HS)

8.3 Resistance to soybean downy mildew

The resistance of soybean plants to the frost mold pathogen *Peronospora manshurica (* Aoum.) Syd.).

0 Immunity (I)

1 highly resistance(HR)

3 Anti-(R)

5 moderate resistance(MR)

7 Sensation (S)

9 High Sensitivity (HS)

8.4 Resistance to purpura

The resistance of soybean seeds to the purple spot pathogen Cecrospora *kikuchii* (T. Matsu & Tomoyasu) Gardner.

1 highly resistance(HR)

3 resist(R)

5 moderate resistance(MR)

7 feel(S)

9 High Sensitivity (HS)

8.5 Resistance to bacterial spot disease

The resistance of soybean leaf blades to the bacterial spot pathogen (*Pseudomonas syringae* pv. *glycine* a) varies in intensity.

1 highly resistance(HR)

3 resist(R)

5 moderate resistance(MR)

7 feel(S)

9 High Sensitivity (HS)

8.6 Resistance to soybean mosaic virus disease

Resistance of soybean leaf blades to soybean mosaic virus (SMV).

0 immunity(I)

1 highly resistance(HR)

3 resist(R)

7 feel(S)

9 High Sensitivity (HS)

8.7 Resistance to Phytophthora sojae root rot

The resistance of soybean plants to Phytophthora sojae Kauf. et Gerde.

3 resist(R)

5 moderate resistance(MR)

7 feel(S)

8.8 Resistance to soybean cysticercosis

The resistance of soybean plants to cyst nematode (Heterodera glycines Ichinohe) disease.

0 immunity(I)

1 highly resistance(HR)

3 resist(R)

7 feel(S)

9 High Sensitivity (HS)

8.9 Resistance to soybean leaf miners

Soybean seeds are susceptible to the soybean weevil (Leguninivora glycinivorella (Mats.obraztsov)) has a weaker resistance.

1 highly resistance(HR)

3 resist(R)

5 moderate resistance(MR)

7 feel(S)

9 High Sensitivity (HS)

8.10 Resistance to soybean aphids

The resistance of soybean plants to aphids (*Aphis glycines* Matsumura)

1 highly resistance(HR)

3 resist(R)

5 moderate resistance(MR)

7 feel(S)

9 High Sensitivity (HS)

8.11 Resistance to pod borers

The resistance of soybean plants to the bean pod moth (*Etiella zinckella (* Treitschke)).

1 highly resistance(HR)

3 Anti-(R)

5 moderate resistance(MR)

7 feel(S)

9 High Sensitivity (HS)

8. 12 Resistance of soybean stem black fly

The resistance of soybean plants to the soybean black stem fly *(Melanagromgza soja Zehntner*).

1 High resistance (HR)

3 Anti-(R)

5 moderate resistance(MR)

7 feel(S)

9 High Sensitivity (HS)

8.13 Resistance to foliar pests

Leaves of soybean plants are susceptible to leaf-feeding pests (Leaf-feeding insect,Insect defoliator Pe-rhaps

The resistance of foliar-feeding insects.

1 highly resistance(HR)

3 resist(R)

5 moderate resistance(MR)

7 feel(S)

9 High Sensitivity (HS)

9 Other Characteristics and Features

9.1 Fingerprint Identification Number

Numbering of soybean germplasm resources in the fingerprint database.

9.2 Core Germplasm Identification

The numbering of soybean germplasm within the core germplasm collection.

9.3 Notes

A detailed description of the specific descriptors or codes for soybean germplasm.